

White Paper 6 — Preparing the purchasable screening library

Pipeline step: Prepare purchasable screening library (ZINC22) **Inputs:** RORyt (ChEMBL1741186) bioactivity data **Outputs:** `screening_library.csv`, `screening_library_3d.sdf`, `fig_library_properties.png`, `LIBRARY_README.md`

1. Purpose

A virtual screen needs a library of candidate molecules. This step assembles a **RORyt-focused, drug-like, 3D-ready** compound deck and documents an important deviation: the intended ZINC22 purchasable source was unavailable, so the library was sourced from ChEMBL bioactivity space instead — with purchasability deferred to a resolver script.

2. Design intent and the ZINC22 outage

The original plan was a purchasable library from **ZINC22** — a catalog of commercially available compounds. During execution, the ZINC22/CartBlanche22 similarity and random-sampling endpoints timed out persistently; only the exact-ID lookup responded. Rather than block the pipeline, we pivoted to a **target-focused bioactivity library** and shipped a `resolve_purchasability.py` script (in the docking package) to annotate purchasability once the backend recovers. This substitution is stated openly in every downstream artifact.

3. Sourcing from ChEMBL bioactivity space

The library was built from **ChEMBL** actives against the RORyt target (ChEMBL1741186) ^[1]. We pulled 714 distinct compounds with measured potency (**pChEMBL** ≥ 7 , i.e. ≤ 100 nM) ^[2], fetched SMILES for the top 250 by potency plus similarity analogs, giving 251 unique structures. A bioactivity-sourced deck has a decisive advantage for a demonstration screen: it contains **built-in positive controls** (known binders), so the screen's ability to recover them is directly testable.

4. Standardization and drug-likeness filtering

Raw database structures were standardized with **RDKit** ^[3] — taking the largest organic fragment and neutralizing charges — then filtered for drug-likeness with a Lipinski-style rule set: at most one Ro5 violation, molecular weight 250–600 Da, ≤ 12 rotatable bonds, and TPSA ≤ 160 Å². This removed non-drug-like outliers and left **151 compounds** (QED 0.28–0.85, MW 278–588 Da).

5. 3D conformer generation

Docking requires 3D structures. Each molecule was embedded with RDKit's **ETKDG** algorithm — Experimental-Torsion Knowledge Distance Geometry, which biases distance-geometry embedding toward torsion angles observed in crystallographic data — followed by MMFF force-field minimization ^[3]. All 151 compounds embedded successfully (seed fixed for reproducibility), producing `screening_library_3d.sdf`.

6. Library properties

The final deck has median MW 479 Da, cLogP 4.5, QED 0.53, and TPSA 72 Å² — a lead-like-to-drug-like distribution appropriate for an LBD pocket. The property distributions are plotted in `fig_library_properties.png`.

7. Deliverables

- `screening_library.csv` — 151 compounds with descriptors and ChEMBL provenance.
- `screening_library_3d.sdf` — 3D conformers ready for docking.
- `fig_library_properties.png` — MW/logP/QED/TPSA and related distributions.
- `LIBRARY_README.md` — full provenance, the ZINC22 outage, and the deferred purchasability plan.

8. Caveats

- The library is **bioactivity-sourced, not confirmed-purchasable**; purchasability resolution is pending ZINC22 recovery (resolver script provided).

- A target-focused deck is ideal for method validation but is not an unbiased chemical-space sample; the full-scale docking package supports substituting a larger purchasable library.

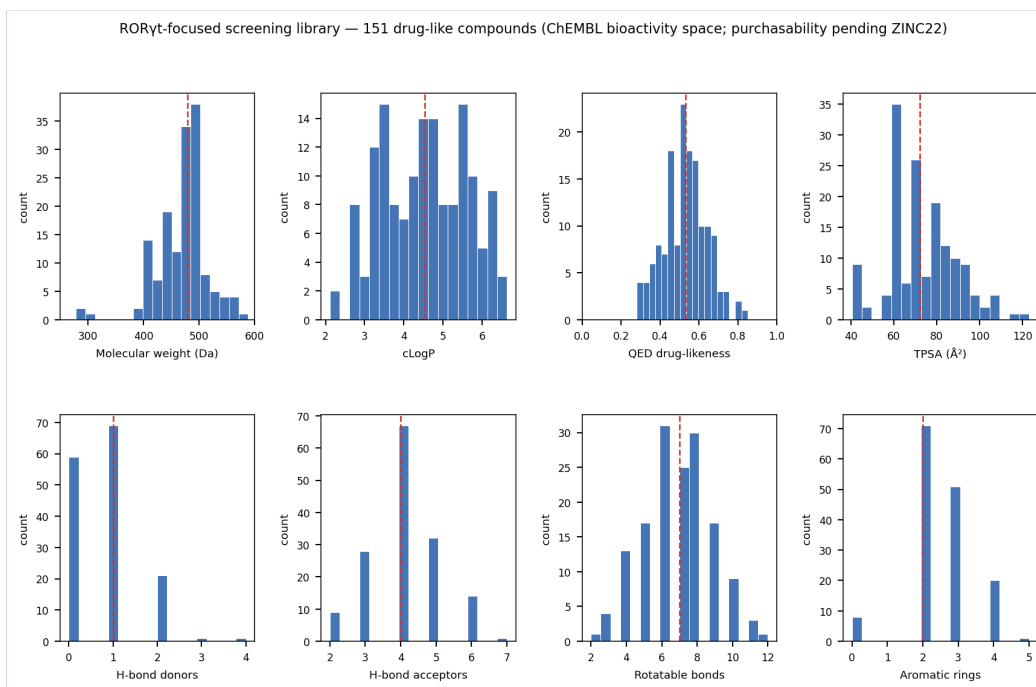


Figure: key output of this pipeline step.

References

1. Zdrazil et al. The ChEMBL Database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*. 2023. doi:[10.1093/nar/gkad1004](https://doi.org/10.1093/nar/gkad1004)
2. Gaulton et al. ChEMBL: a large-scale bioactivity database for drug discovery. *Nucleic Acids Research*. 2011. doi:[10.1093/nar/gkr777](https://doi.org/10.1093/nar/gkr777)
3. Riniker et al. Better Informed Distance Geometry: Using What We Know To Improve Conformation Generation. *Journal of Chemical Information and Modeling*. 2015. doi:[10.1021/acs.jcim.5b00654](https://doi.org/10.1021/acs.jcim.5b00654)